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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Savings Goal Planning and Financial Progress Tracking System

Submitted by

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APRIL 2026

Semester 2

DECLARATION

I hereby declare that the project entitled "**Savings Goal Planning and Financial Progress Tracking System"** submitted in partial fulfillment of the requirements for the course Java Programming is a **record of my original work** carried out under the guidance of **[Trainer Name]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

HABI E

CERTIFICATE

This is to certify that the project entitled **"Savings Goal Planning and Financial Progress Tracking System"** is a **bona fide work** carried out by **HABI E, 1st Bsc CS 'A'**, B.Sc.Computer Science, **KG College of Arts &**

Science, in partial fulfillment of the requirements for the course **Java Programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: KGCollege of Arts and Science

Date: 10.4.2026

Submitted for the Viva-Voce Examination held on _____10.4.2026_____

Guide: **Head of the Department**(Technical Trainer, KGCAS) (Technical Head, KGCAS)

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ABSTRACT

The **Savings Goal Planning and Financial Progress Tracking System** is a JavaFX-based application designed to help users manage and track their savings efficiently. Many individuals find it difficult to monitor their savings progress manually, which may lead to poor financial planning.

This project provides a simple solution where users can set a savings goal, add savings amounts, and track their progress visually using a progress bar. The system calculates the remaining amount and percentage of completion automatically.

The application is developed using Java and JavaFX, applying programming concepts such as event handling, conditional statements, and exception handling. The main objective of the project is to provide a user-friendly tool for personal financial management and to demonstrate practical implementation of Java GUI applications.

1. INTRODUCTION

In the hospitality industry, managing room reservations efficiently is important for smooth operations and customer satisfaction. In many small hotels, room booking is done manually, which can lead to problems such as double booking, confusion in records, and difficulty in tracking guest information.

Manual systems are time-consuming and prone to human errors. Staff may accidentally assign the same room to multiple guests or fail to update records properly during check-out.

With the advancement of technology, automated reservation systems have become essential. A computerized system can manage bookings accurately, prevent conflicts, and maintain proper records.

The Hotel Room Reservation and Guest Management System is developed using JavaFX to provide a simple and effective solution for managing hotel bookings. The system allows users to enter guest details, reserve rooms, check room availability, and perform check-out operations.

This project demonstrates the use of Java programming and GUI development in solving real-world problems in hotel management.

1.1 Objective

- To develop a JavaFX-based savings tracker
- To allow users to set savings goals
- To track savings progress visually
- To calculate remaining amount automatically
- To demonstrate Java GUI and event handling concepts

1.2 Overview

The system is a GUI-based application that allows users to enter a savings goal and update their savings regularly.

The system checks for valid input and updates savings details dynamically. It also displays progress using a progress bar for better visualization

1.3 Features

The **SavingsGoalPlanningandFinancial Progress Tracking System** includes the following main features:

- Set Savings Goal
- Add Savings Amount
- View Saved Amount
- Calculate Remaining Amount
- Display Progress Percentage
- Progress Bar Visualization

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

Many people struggle to track their savings manually, leading to:

Lack of financial discipline

Calculation errors

No clear progress tracking

2.2 Project Overview

The Savings Goal Tracker provides:

Input for goal setting

Input for adding savings

Automatic calculations

Visual progress tracking

2.3 Scopes and Limitations

Scope:

Personal use

Simple savings tracking

Demonstration of JavaFX concepts

Limitations:

No database (temporary storage only)

Single user system

Basic interface

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

Input Module – Accepts goal and savings input

Processing Module – Performs calculations

Storage Module – Stores values in variables

Output Module – Displays results

Progress Module – Updates progress bar

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Savings Goal Planning and Financial Progress Tracking System,** a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis• System Design• Program Development• Testing and Validation• Result and Documentation

4.1. Problem Analysis

Understanding the need for a savings tracking system and identifying user requirements.

4.2. System Design

Designing the GUI layout and logic flow using JavaFX components..

4.3. Program Development

Creating UI components

Writing event handling code

Implementing logic for calculations

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Testing with:

Valid inputs

Invalid inputs

Edge cases

Ensures correct working of the application.

4.5. Result and Documentation

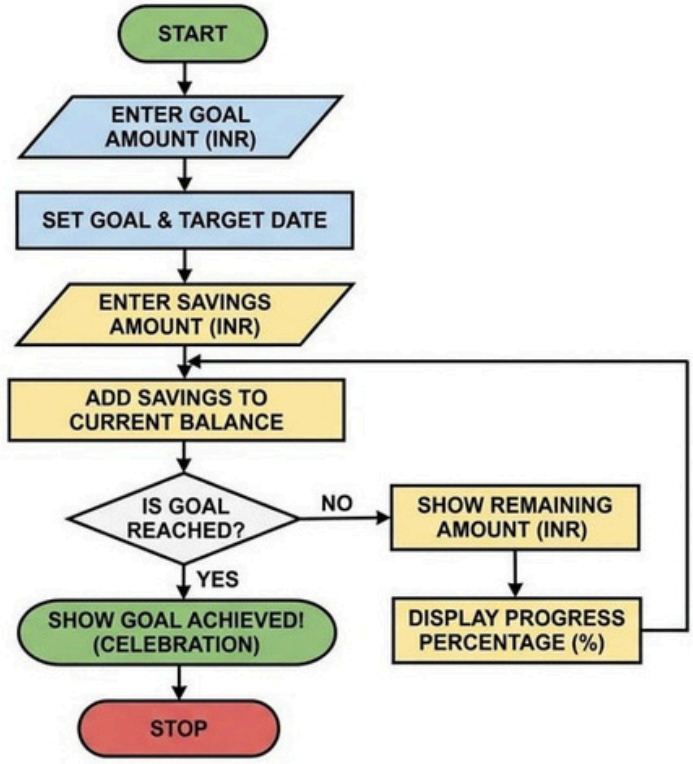
The system works successfully and displays accurate results. Output is shown through GUI labels and progress bar.

4.6. Algorithm

Algorithm for the **Savings Goal Planning and Financial Progress Tracking System:**

1. Start
2. Enter goal amount
3. Check validity
4. Set goal
5. Enter savings amount
6. Check if goal set
7. Add savings
8. Calculate remaining
9. Calculate progress
10. Display results
11. Update progress bar
12. End

4.7. Flowchart



5. SOURCE CODE

```
import javafx.application.Application;
import javafx.geometry.Insets;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.layout.*;
import javafx.stage.Stage;

public class App extends Application {

    double goalAmount = 0;
```

```
double savedAmount = 0;

@Override
public void start(Stage stage) {

    //      Title
    Label title = new Label("Savings Goal Tracker");
    title.setStyle("-fx-font-size: 18px; -fx-font-weight: bold;");

    //      Goal Input
    TextField goalField = new TextField();
    goalField.setPromptText("Enter Savings Goal");

    Button setGoalBtn = new Button("Set Goal");

    Label goalLabel = new Label("Goal: Not set");

    setGoalBtn.setOnAction(e -> {
        try{
            goalAmount = Double.parseDouble(goalField.getText());
            goalLabel.setText("Goal: ₹" + goalAmount);
            goalField.clear();
        }catch (Exception ex) {
            showAlert("Enter valid number!");
        }
    });

    //      Add Savings
    TextField addField = new TextField();
    addField.setPromptText("Add Savings Amount");
```

```
Button addBtn = new Button("Add Savings");

Label savedLabel = new Label("Saved: ₹0");
Label remainingLabel = new Label("Remaining: ₹0");
Label progressLabel = new Label("Progress: 0%");

ProgressBar progressBar = new ProgressBar(0);

addBtn.setOnAction(e -> {
    try{
        double amount = Double.parseDouble(addField.getText());

        if(goalAmount == 0) {
            showAlert("Set goal first!");
            return;
        }

        savedAmount += amount;

        double remaining = goalAmount - savedAmount;
        double progress = savedAmount / goalAmount;

        savedLabel.setText("Saved: ₹" + savedAmount);
        remainingLabel.setText("Remaining: ₹" + Math.max(remaining, 0));
        progressLabel.setText("Progress: " + (int)(progress * 100) + "%");

        progressBar.setProgress(Math.min(progress, 1));

        addField.clear();
```

```
    }catch (Exception ex) {
        showAlert("Enter valid number!");
    }
});

//      Layout
VBoxgoalBox = new VBox(10,
        new Label("Set Savings Goal"),
        goalField,
        setGoalBtn,
        goalLabel
);

goalBox.setPadding(new Insets(10));
goalBox.setStyle("-fx-border-color: black;");

VBoxtrackBox = new VBox(10,
        new Label("Track Progress"),
        addField,
        addBtn,
        savedLabel,
        remainingLabel,
        progressLabel,
        progressBar
);

trackBox.setPadding(new Insets(10));
trackBox.setStyle("-fx-border-color: black;");
```

```
        HBox mainLayout = new HBox(20, goalBox, trackBox);
        mainLayout.setPadding(new Insets(10));

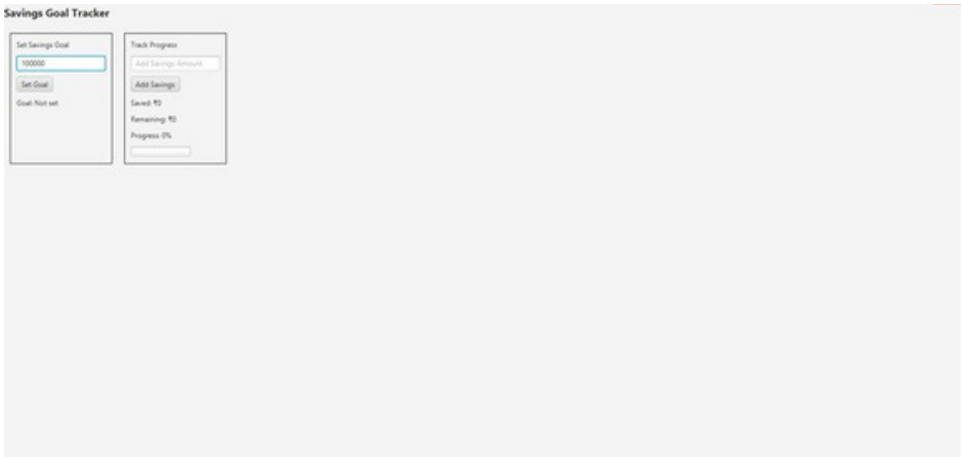
        VBox root = new VBox(10, title, mainLayout);

        Scene scene = new Scene(root, 600, 400);
        stage.setScene(scene);
        stage.setTitle("Savings Tracker");
        stage.show();
    }

    //    Alert method
    private void showAlert(String msg) {
        Alert alert = new Alert(Alert.AlertType.ERROR);
        alert.setTitle("Error");
        alert.setHeaderText(null);
        alert.setContentText(msg);
        alert.showAndWait();
    }

    public static void main(String[] args) {
        launch();
    }
}
```

6. INPUT AND OUTPUT SCREENS





7. DISCUSSION & CONCLUSION

7.1 Discussion

Thesystemwas developed using JavaFX to automate hotel reservations. It successfully manages bookings, prevents double booking, and handles guest check-out efficiently.

The system was developed using JavaFX and successfully tracks savings progress. It uses core programming conceptslikeconditions, exception handling, and GUI design.

7.2 Conclusion

The **Savings Goal Planning and Financial Progress Tracking System** is a simple and effective tool for managingpersonalsavings.Itreducesmanualeffortandhelpsusersachieve their financial goals efficiently.

8. REFERENCES

The following resources were used during the development of this project:

Java: The Complete Reference – Herbert Schildt

Oracle Java Documentation

GeeksforGeeks

W3Schools